Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/04/2014
PI Name:	Wotring, Virginia Ph.D.		
Project Title:	Effect of Spaceflight on Expression of Metabolic Enzyme Genes in Mice		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Virginia.Wotring@bcm.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	Baylor College of Medicine		
PI Address 1:	Center for Space Medicine		
PI Address 2:	6500 Main St, Suite 910		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	9
Comments:	PI formerly with Universities Space Research Association until fall 2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	06/23/2011	**End Date:**	09/30/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:	Shuttle		
Flight Assignment:	STS-135 tissue sharing		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	Most pharmaceuticals are metabolized by the liver, and clinically-used medication doses are given with normal liver function in mind. Metabolism, specifically the metabolism of administered medications, is a large part of pharmacokinetics and among the chief determinants of concentration and duration of action of administered medications. Because of this, metabolism can make the difference between a drug that works, and one that doesn't work or causes side effects. Metabolism is also highly variable; it can be influenced by an individual's genetic makeup and by a number of environmental factors. Exposure to gamma-irradiation is one of these factors; it has been shown to alter expression of many genes including some that metabolize drugs. Elevated exposure to gamma-irradiation is experienced during spaceflight. The purpose of this study is to measure the effect of spaceflight radiation on liver gene and protein expression, in order to learn more about potential effects of spaceflight on drug metabolism. A recent JSC HRP-funded pilot study entitled		

Task Description:	"Radiation Damage in Organ Tissues of Mice" allowed examination of a large number of genes involved in drug metabolism and DNA repair. There were several compelling (preliminary) findings. Expression of genes involved in metabolism of steroid hormones and drugs with similar structures was elevated, potentially important because several medications used in flight have this kind of structure. Many antioxidant genes were also upregulated; one (metallothionein) increased ~200-fold in mice exposed to 6 Gy at 4 h post exposure. At 24 h after exposure, metallothionein expression was still elevated compared to controls, but only about 5-fold. The stimulus in this pilot was a single high radiation dose (6 Gy) delivered acutely, perhaps not the best model for radiation exposure typically encountered in spaceflight. This new proposal is to perform exactly the same analyses on animals that have been exposed to spaceflight, allowing direct comparison of results with the previous pilot study. This study will investigate gene and protein expression in response to exposure to the spaceflight environment for 12 days. Expression of a select group of genes would first be examined in liver tissue from each group of animals, spaceflight and ground control. In a second phase, gene products (proteins) would be examined for those genes that show significant expression changes. It is not known whether the gene expression changes measured in the previous pilot study are followed by changes in enzyme expression or activities (nor has this been demonstrated adequately in the literature); the protein experiments in this proposal would begin to address that question. Specific Aim 1: To determine the effect of short duration spaceflight on expression of liver genes involved in oxidative stress and repair and those involved in drug metabolism. Specific Aim 2: In affected genes, determine relationships between protein expression and gene expression. Specific Aim 3: To compare results from Aims above with the results of previous pilot study of liver gene expression after ground exposure to gamma irradiation ("Pilot Studies of Radiation Damage in Organ Tissues of Mice", Wu & Wotring).
Rationale for HRP Directed Research:	The proposed pilot/feasibility study is requesting the use of spaceflight animal tissues from STS-135 that may be made available by the NASA ISS Research Project (ISSRP) for tissue sharing studies. There is insufficient time for ISSRP or HRP to solicit for this near-term opportunity. ISSRP is considering proposals for tissue sharing that are consistent with, and supplement, existing NASA-funded research grants and/or directed research studies.
Research Impact/Earth Benefits:	
Task Progress:	Results showed significant changes in gene expression in many genes associated with metabolism of administered medications, drug transport across cell membranes, reduction-oxidation homeostasis, and repair of DNA damage. Analysis of expressed protein in the samples showed similar trends.
Bibliography Type:	Description: (Last Updated: 12/24/2019)